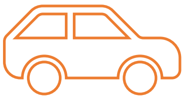
Πλ

11

S HCF (6,72, 120)=2x32x3-6

32, 3 and 5 are the greatest powers of the prime factors 2, 3 and 5 respectively involved in the three

numbers.  

So

1. CM(6, 72, 120) = $2 ^ \gamma * 3 ^ 2 * 5 ^ 1 = 360$.

Remark: Notice, 6 x 72 x 120 HCF (6, 72, 120) x LCM (6, 72, 120). So, the product of three numbers is not equal to the product of their HCF and LCM.

EXERCISE 1.2

1. Express cach number as a product of its priame factors:

(1)140

(156

(8)3825

Shed

2. Find the t 1CM and HCF of the following pairs of integers and verify thui LCMHCF= prodict of the two numbers,

(i) 26 91

510 and 92

(336 and 54

3. Find the LCM and HCF of the following integer integers by applying the prime tacionisanon merhot

1012,15 and 21

(17,23 and 29

(4) 8.9 and 25

4. Given that HCF (306,657)=9, find LCM (306.657)

5. Check whether can end with the digit 10 for any natural number w

6. Explain why 7 * 11 * 13 + 13z 7 * 6 * 5 * 4 * 3 * 2 * 1 + 5z rccomposite numbers

7. There is a circular path around a sports field. Sonia takes 18 minutes to drive one round of the field, while Ravi takes 12 minutes for the same Suppose they both start at the sarve point and at the same time, and go in the same direction. After how many minutes will they meet again at the starting point?

1.4 Revisiting Irrational Numbers

In Class IX, you were introduced to irrational numbers and many of their properties. You studied about their existence and how the rationals and the irrationals together made up the real numbers, You even studied how to locate irrationals on the number line. However, we did not prove that they were irrationals. In this section, we will prove that sqrt(2) sqrt(3) sqrt(5) and, in general, p is irrational, where p is a prime. One of the theoremsour proof, is the Fundamental Theorem of Arithmetic, Recall, a number is called irrational if it cannot be written in the form where p and q are integers und 40. Some examples of irrational numbers, with